

Assignment 1

Q.1) Define the Real and Virtual Memory Concept.

→ Real Memory

- Real memory refers to the actual physical RAM installed in computer system.
- The CPU access real memory directly using physical address to read and write data.
- Real memory has a limited capacity, which can be a constraints for systems running multiple application simultaneously.

Virtual Memory

- Virtual memory is a memory management technique that combines real memory (RAM) with secondary storage (hard disk or SSD) to provide a larger address space.

• When the CPU accesses Virtual memory, it uses virtual addresses that are mapped to physical addresses in real memory or secondary storage.

• Virtual memory uses paging and swapping techniques to move data between real memory and secondary storage, allowing system to run application that require more memory than is physically available.

Q.2) Mention different types of Computer Memory.

→ Types of Computer Memory.

1. RAM (Random Access Memory) :
Volatile memory that temporarily stores data and application while a computer is running.

2. ROM (Read Only Memory) :
Non-Volatile memory that permanently stores data.

3. Cache Memory :
Small, fast memory that stores frequently access data to improve system performance.

4. Virtual Memory :
Combination of RAM and secondary storage that provide a larger address space for application.

5. Primary Memory :
Directly accessible memory that includes RAM and ROM.

6. Secondary Memory :
Non-Volatile storage devices such as hard drives, SSDs and flash drives that store data and programs long-term.

Q3. Discuss the key architectural improvements in modern microprocessors.

→ The key architectural improvement in modern microprocessors.

- Multicore and Manycore Systems: Integrating multiple processing cores on a single chip enables parallel processing, increasing overall system performance and efficiency. This architecture allows for more efficient use of resources, reducing power consumption.

- Pipelining and Super-pipelining: Pipelining breaks down the instruction cycle into stages, enabling multiple instructions to be processed simultaneously. Super-pipelining further segments these stages, increasing clock rates and improving performance.

- Superscalar Architecture: This design allows for the execution of multiple instructions

forms a single instruction stream in parallel, improving instruction-level parallelism and overall system performance.

- Heterogeneous Computing: Integrating different types of processing unit, such as CPUs, GPUs and TPUs on a single chip improves overall system performance.

- Increased Use of Specialized Case Specialized Cores like Graphics Processing Unit (GPU) and Tensor Processing Unit (TPU) have become increasingly important for specific workloads, such as artificial intelligence and machine learning.

8.4 Describe the function of the ALU, CU and registers.

→ Arithmetic Logic Unit (ALU).

- The ALU performs arithmetic operation such as addition, subtraction, multiplication & division

- ALU performs logical operation such as AND, OR and NOT

- ALU processes data according to the instructions provided by control unit.

Control Unit (CU).

- The control unit decodes instruction and generates control signals to execute them

- Control unit manages the flow of data b/w different parts of system such as ALU, registers, & memory.

- Control unit determines the sequence of instruction to be executed

Registers

- Registers are small, high-speed memory location that store data temporarily while it is being processed.

- Register can transfer data to and from ALU, memory and other parts of system.

- Register can also store memory addressing, allowing the CPU to access specific location in memory

8.5. Explain the different types of interrupts

→ Types of Interrupts.

1) Hardware Interrupts :

Generated by hardware devices such as keyboards, mice or disk drives, to signal the CPU that attention is required.

2. Software Interrupts :-

Generated by software instruction, often used to request system services or to handle exception.

3. Maskable Interrupts :-

Can be enabled or disabled by the CPU, allowing for prioritizing and control of interrupts handling.

4. Non-Maskable Interrupts :-

Cannot be disabled by the CPU and are typically used for critical events, such as system crashes or hardware failures.

5. Vectored Interrupts :-

The interrupting devices provides a vector (address) to interrupt handler, to handle interrupting source.

6. Non-Vectored Interrupts :-

The CPU uses a fixed address or routine to handle the interrupt, regardless of interrupting device.

Q 6. Discuss the importance of interfacing devices.

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Enabling Communication :-

Interfacing devices enable communication between different components or systems, allowing them to exchange data, control signals, or power.

Expanding System Capabilities :-

Interfacing device can expand the capabilities of a system by adding new functionality, such as input/output operation.

Ensuring Compatibility :-

It ensures compatibility b/w different system or components, allowing them to work together seamlessly.

Improving System Performance :-

It improves system performance by optimizing data transfer rates, reducing latency.

• Enhancing System Flexibility
It enhance system flexibility by allowing for connection of different devices, peripherals or system.

Q7). Why do you understand by Addressing Mode? Explain with example.

→ Addressing Modes

It determines how CPU accesses and manipulate data in memory. They specify the way CPU calculates the effective address of operand.

Types of Addressing Modes

1. Immediate Addressing Mode :

The operand is specified directly in the instruction.
Example: `MOV AX, 5` (loads the value 5 into AX register).

2. Register Addressing Mode :

The operand is stored in a register.
Example: `MOV AX, BX` (loads

the values in BX register into AX register).

3) Direct Addressing Mode :

It specifies the memory address of operand.
Example: `MOV AX, [1000H]` (loads value at memory Address 1000H into AX register).

4) Indirect Addressing Mode

The instruction specifies a register as memory location that contains memory address of operand.
Example: `MOV AX, [BX]`

5) Indexed Addressing Mode :

It specifies a base address that is used to calculate effective address.
Example: `MOV AX, [BX+5]`

Q8) State various Microprocessor Development Schemes.

→ Microprocessor Development Schemes

1. Pipelining: Breaks down instruction cycle into stages, enabling multiple instruction to be processed simultaneously.
2. Superscalar Architecture: Allows for the execution of multiple instruction from a single instruction stream in parallel.
3. Multi-core Processing: Integrates multiple threads to share same processing resources.
4. Cache Hierarchy: Uses multiple levels of cache memory to optimize data access and reduce latency.
5. Branch Prediction: Predicts the outcome of branch instruction, reducing the overhead of mispredicted branches.

6) Instruction-Level ~~Parallelism~~
Parallelism (ILP) ◦ Exploits
parallelism at instruction level,
improving performance and
efficiency.